**Solar ESS Co. Ltd.**
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
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SOLAR + BATTERY GUIDE

# How long can batteries run your air conditioners?

A simple guide for 6 air cons on 4 × 16 kWh LiFePO4 batteries

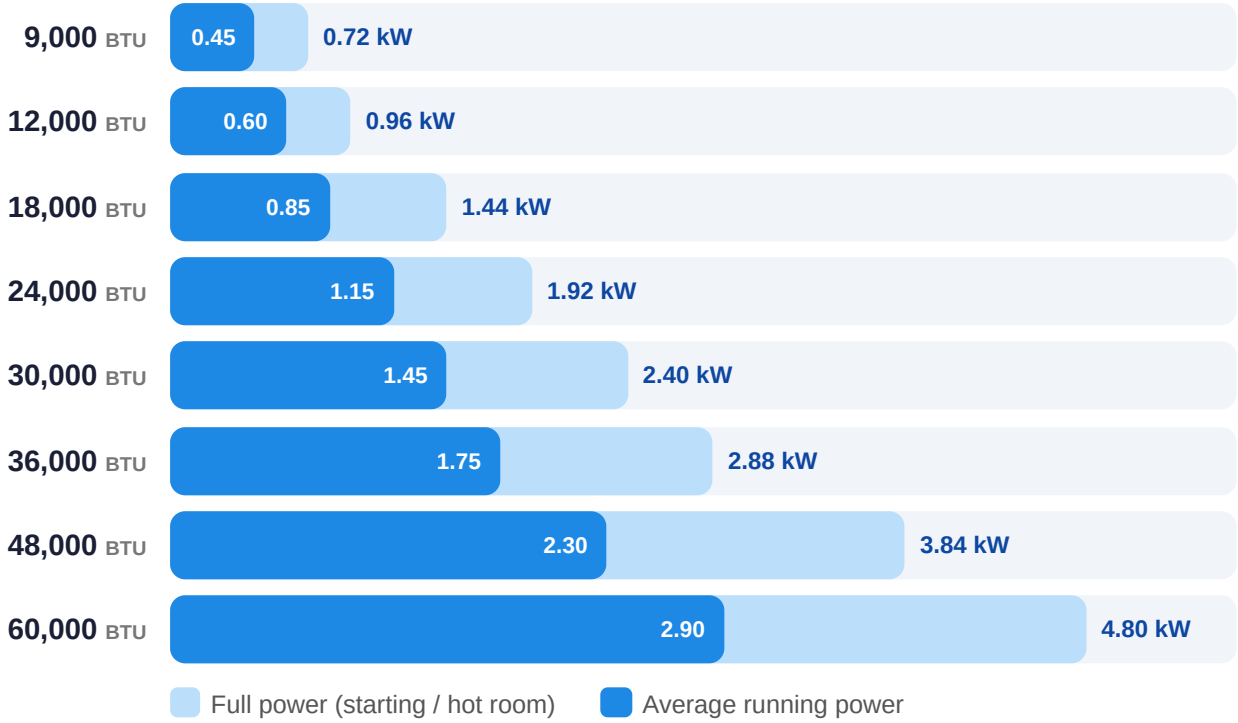
❄️ 6 air cons

🔋 4 × 16 kWh

⏱️ hours of cooling

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# 1 How much power does an air con use?




Figures for inverter air cons (EER ≈ 12.5). Non-inverter units use about 15% more.

Rule of thumb

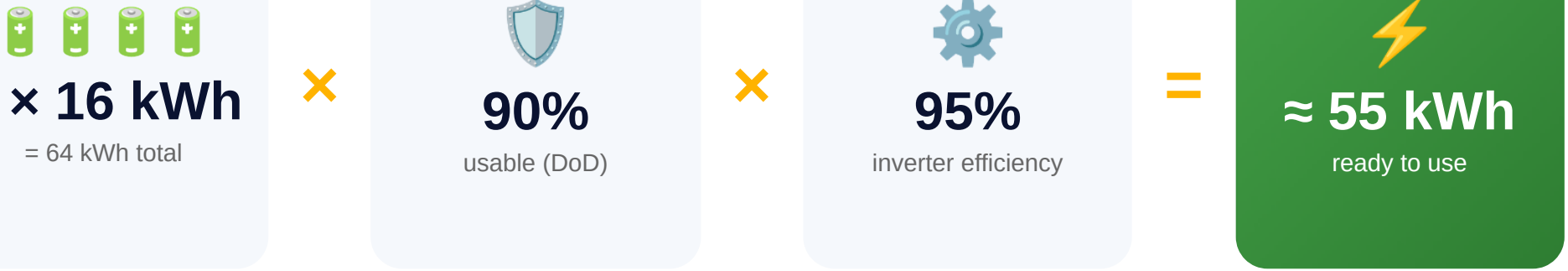
**12,000 BTU ≈ 1 kW**

Power (kW) = BTU ÷ (EER × 1,000)

Inverter air cons slow down once the room is cool, so on average they use only **about 60%** of full power.

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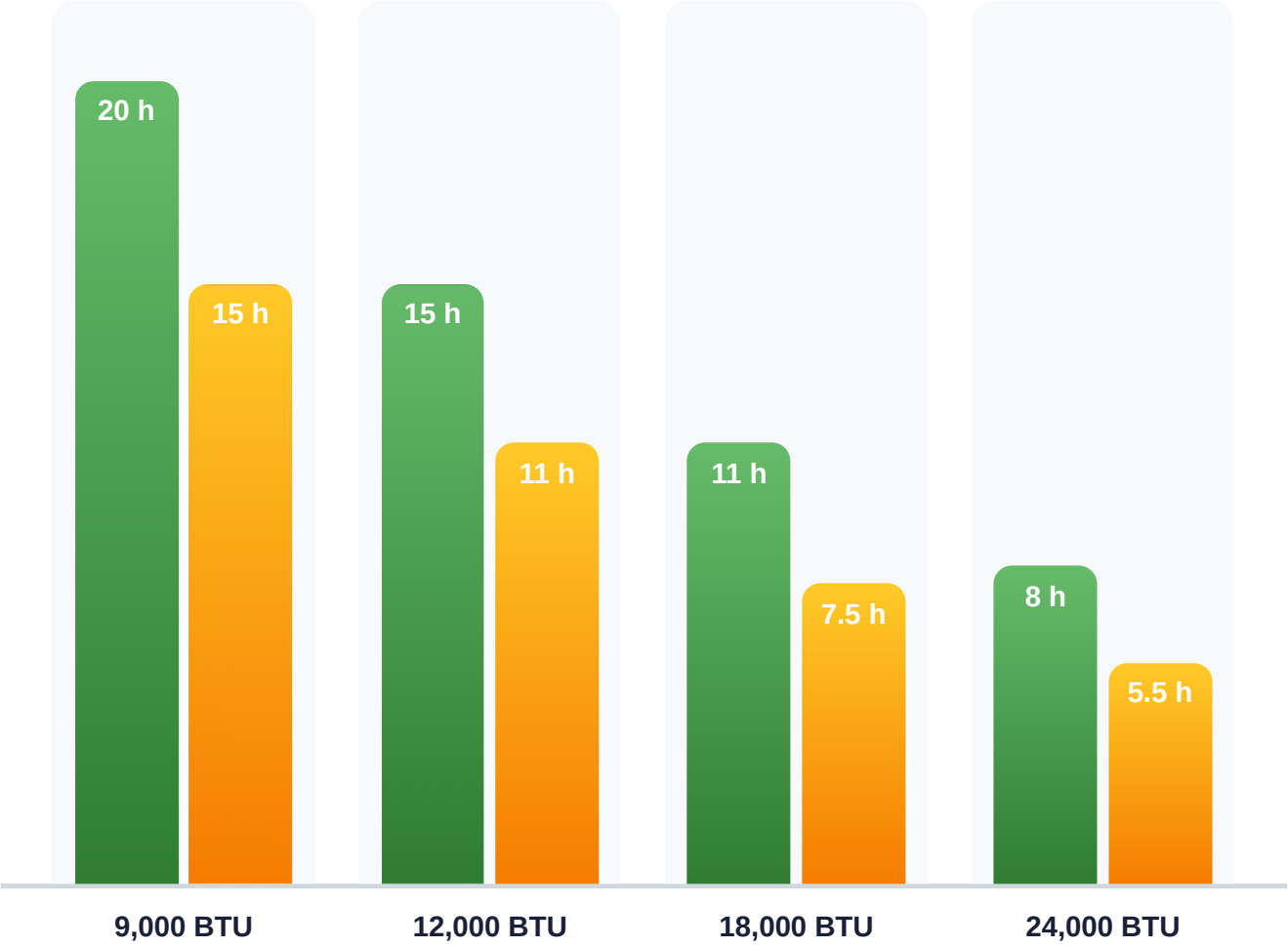
## 2 How much energy do 4 batteries hold?




💡 **55 kWh ≈ 55 units of electricity**, the same "units" shown on your PEA bill.

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# 3 How many hours for 6 air cons?




Hours = 55 kWh ÷ (6 × kW per air con)

Inverter air con  Non-inverter air con

6 × 12,000 BTU inverter

**≈ 15 hours**

enough for the whole night

## 4 Tips for the longest runtime



### Set 25–26 °C

Every degree cooler adds roughly 6–8% more power use.



### Choose inverter air cons

About 25–35% longer runtime than non-inverter units.



### Start them one at a time

Air cons draw a surge when starting. Staggering protects the inverter.



### Size the inverter too

6 × 12,000 BTU ≈ 6 kW → one 10 kW inverter is OK. 18,000–24,000 BTU needs two inverters.

☀️ Solar ESS designs the right battery and inverter size for your home or business · Tel 063 151 6694